

# Hailo-8L NPU – Technical Description

## 1. What Is a Neural Processing Unit (NPU)?

A Neural Processing Unit (NPU) is a specialized processor designed specifically for deep-learning inference workloads. Instead of being a general-purpose engine like a CPU or a graphics-oriented engine like a GPU, an NPU focuses on neural-network operations such as convolutions, matrix multiplications, activation functions, and tensor movement. By specializing the hardware around these operations, NPUs can achieve very high throughput and low power consumption, especially in edge devices.

## 2. Differences Between CPU, GPU, and NPU

Instead of a dense grid, this section describes each type of processor in its own short subsection for readability.

### 2.1 CPU (Central Processing Unit)

- General-purpose processor used for operating systems, applications, and control logic.
- Contains a small number of powerful cores with deep cache hierarchies.
- Optimized for complex branching, sequential code, and low-latency single-thread tasks.
- Can run AI workloads, but is usually inefficient for large neural networks.

### 2.2 GPU (Graphics Processing Unit)

- Originally designed for graphics rendering and image processing.
- Contains many simple cores arranged for massively parallel floating-point math.
- Very effective for training and inference of deep neural networks, but still relies heavily on off-chip memory bandwidth and a general-purpose compute model.
- High throughput but also relatively high power consumption for continuous AI workloads.

### 2.3 NPU (Neural Processing Unit)

- Purpose-built for neural-network operations, such as convolutions, fully connected layers, and activation functions.
- Includes dedicated tensor hardware and on-chip memory fabric to minimize data movement.
- Provides the highest performance per watt for inference, especially in edge or embedded systems.
- Executes neural networks as optimized dataflow graphs rather than generic instruction streams.

### 3. What Is the Hailo-8L NPU?

The Hailo-8L is a low-power, PCI Express (PCIe)-based AI accelerator from Hailo, designed primarily for embedded edge devices such as single-board computers (e.g., Raspberry Pi 5), smart cameras, IoT gateways, drones, and robots.

Key characteristics include:

- A dedicated neural-network acceleration engine based on Hailo's Dataflow Architecture.
- Up to roughly 13 TOPS (tera operations per second) of AI compute, depending on the model and configuration.
- Very high performance per watt, typically consuming only a few watts during inference.
- Tight integration with the Hailo software stack, including HailoRT (runtime), the Hailo Dataflow Compiler, and the HEF (Hailo Executable File) model format.

On platforms like the Raspberry Pi 5, the Hailo-8L offloads the heavy tensor computations from the CPU and GPU. The Pi handles control, pre/post-processing, and I/O, while the NPU executes the core neural network layers.

### 4. Hailo NPU Architecture (Dataflow Architecture)

The Hailo-8L is built around a dataflow-centric architecture. Instead of executing instruction streams like a CPU or massive parallel threads like a GPU, Hailo compiles the neural network into a hardware-execution graph that runs across specialized compute units and on-chip memory.

#### 4.1 Key Architectural Components

- **Tensor Blocks (TBs):** Specialized compute units that execute operations such as convolutions, fully connected layers, pooling, and activations. Each Tensor Block has its own local SRAM.
- **Control Units / Scheduler:** The Hailo Dataflow Compiler transforms a high-level neural network into a hardware execution graph. A scheduler orchestrates which Tensor Blocks are active and how data flows between them.
- **On-Chip Distributed SRAM:** SRAM banks are placed close to compute units to keep tensor data local, reducing latency and external memory access.
- **Data Movement Fabric:** A dedicated routing fabric connects Tensor Blocks and SRAM banks, allowing tensors to move efficiently from one operation to the next without relying on global DRAM traffic.
- **DMA Engines / PCIe Interface:** The host streams input tensors to the NPU and receives output tensors back over the PCIe link.

#### 4.2 How Hailo Executes a Neural Network

1. A trained model (for example, in ONNX format) is compiled with the Hailo Dataflow Compiler, producing an HEF file.
2. The compiler partitions the network into subgraphs and maps layers onto specific Tensor Blocks, SRAM banks, and routing paths in the fabric.
3. At runtime, the host application uses HailoRT to load the HEF, configure the device, and set up input/output streams.

4. Input tensors (such as image frames) are sent to the NPU, which processes them layer by layer as the data flows through the mapped Tensor Blocks.
5. Final outputs (for example, bounding boxes and class scores for an object detector) are returned to the host for post-processing, visualization, or control logic.

### 4.3 Conceptual Architecture Diagram

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| \-----+-----+-----/ | | On-Chip Distributed SRAM | | | Control / Scheduler Unit | | | PCIe  
Interface | +-----+

### 5. Summary

The Hailo-8L NPU is a dedicated neural-network accelerator that uses a dataflow architecture to map neural networks directly onto specialized compute blocks and on-chip memory. Compared with CPUs and GPUs, this approach minimizes unnecessary data movement, reduces latency, and dramatically improves performance per watt for inference workloads. On platforms such as the Raspberry Pi 5, the Hailo-8L enables high-frame-rate, low-power AI applications while leaving the main CPU and GPU free for other tasks.